Measurements with highspeed ultrasound in different materials and with different applications

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ABSTRACT

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1 Introduction

Ultrasound describes the frequencies of sound waves, that are so large such that the human ear can not hear them. According it's inaudibility for the human, ultrasound was not utilized until late. Today ultrasound is used for spectroscopic examinations of materials and detections of objects, where other methods would be either to invasive (namely ultrasound in the medicin/materials science) or not precise enough (e.g. sonar).
In the following the propagation behavior of ultrasound in different materials will be examined, as well as measurements utilizing the properties of ultrasound.

2 Basic functions of an echoscope

An echoscope is an ultrasonic measurement system that connects a probe to a computer or an oscilloscope. The probe is commonly made of a transducer in form of a piezocristal. The echoscope then gives then an electical signal with a frequency to the probe which causes the piezocristal to vibrate and therefore it emits a supersonic wave. The probe not only emits ultrasound but it can act as a detector aswell when the piezocristal receives a vibration and converts it into an electical signal which then can be amplified and measured by the echoscope.
There are two main meathods of measurements using a echoscope: Firstly one probe emits a wave and another probe receives the incoming wave. Here the two probes have to me aligned perfectly to not get disturbances. Secondly one probe emits a wave and the same probe also receives it using the reflection at the end surface. The probe has to be aligned perpendicular to the end surface to not get disturbances.
There are multiple ways to process incoming signals from the echoscope. In this paper we want to highlight two. The most important one is the A-scan (amplitude scan) where the probe is kept at a constant position, measuring the amplitude of the emitted and received signal depending on the time after a pulse has been sent by the transmitter. Another way one can use an echoscope is the B-scan(brightness scan), where the probe is moved over an object and a 2D-Image of the plane in which teh probe was moved in is created. The amplitudes are given as different brightness values.

3 Experiment

3.1 Velocity of ultrasound in polyacryl

To measure the propagation speed of sound waves through media, three polyacryl cylinders with different lenghts were used. The time that the wave took to travel through the cylinder was measured via the method of reflection, where the ultrasound transmitter also worked as the detector of the signal. The time was then captured with the echoscope. With the method of reflection one has to notice that the length in the formula for the propagation speed is no double the length of the cylinder, concluding in:

$$v_{poly} = \frac{2 \cdot l}{\Delta t} \tag{1}$$

As noted above l is the total length of the cylinder, Δt is the time the wave took to travel twice through the cylinder and v_{poly} is the resulting speed of sound. This method was then applied to all three different cylinder-lengths, each with three different frequencies $f_{1MHz} = 1MHz$, $f_{2MHz} = 2MHz$ and $f_{4MHz} = 4MHz$. Since the absorption for f_{4MHz} was big, the largest cylinder could not be measured.

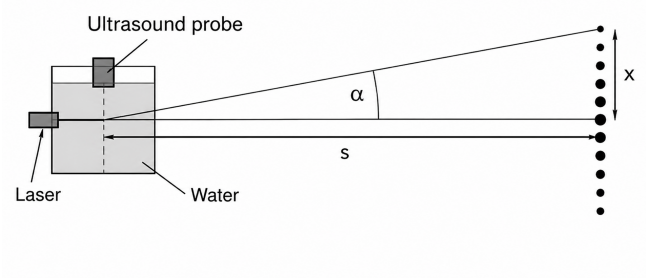

Figure 1: Setup used to measure the wavelength of ultrasound utilizing the Dabye-Sears effect

With the relevant data shown in 5.1, Tab. 1 the mean sound velocitys for each frequency is calculated:

$$v_{poly.1MHz} = (2.619 \pm 0.032)\frac{km}{s}$$
$$v_{poly.2MHz} = (2.677 \pm 0.033)\frac{km}{s}$$
$$v_{poly.4MHz} = (2.746 \pm 0.052)\frac{km}{s}$$

It is clearly visible, that the speed of sound in polyacryl depends on the frequency of the wave traveling through it. Thus it indicates dispersion in the sound velocity.
In the data sheet from a manufacutrer, the longitunal wave velocity for the frequency of 1Mhz is given by $v_{poly,Lit} = 2.7\frac{km}{s}$ [3], therefore the measured value does not match the literature within the error. This could be due to the fact that the measurement was done in saturation of the used piezoelectric ultrasound emitter. Also because of the absorption of the material, the gain on the echoscope had to be increased, which also increased the noise of the measurement thus increasing the uncertainty.

3.2 Absorption of ultrasound in PVC

3.3 Measurements on a polyacrylic test block

3.4 Resolution of an echoscope

3.5 Spectral methods of an echoscope

3.6 Debye-Sears effect

The Dabye-Sears effect describes an acousto-optic effect, where the ultrasound probe creates a standing wave with lattice constant $g = \lambda_S$ perpendicular to an incoming laser beam. The beam then diffracts as it does with a classical optical lattice. With the formula for the lattice spectrum from classical optics one can determine the wavelength λ_S of the ultrasound. The setup of the measurement is shown in fig. 1.

$$\lambda_S = \frac{n \cdot \lambda_{laser}}{sin(\alpha)} \overset{(1)}{\approx} \frac{n \cdot \lambda_{laser} \cdot s}{x} \tag{2}$$

In eq. 2 n is the number of the maxima, λ_{laser} is the wavelenght of the laser and α is the angle of diffraction. The approximation (1) in eq. 2 follows from the large difference in distance $x \ll s$.

4 Conclusion

References

[1] `https://www.physik.uni-wuerzburg.de/fileadmin/11000000/03_Studium/02_Bachelor/Grundpraktikum/_imported/fileadmin/11016800/Modulbeschreibungen/C-Modul/V35.pdf` last visited on 06.05.2026

[2] `https://www.physik.uni-wuerzburg.de/fileadmin/11000000/03_Studium/02_Bachelor/Grundpraktikum/_imported/fileadmin/11016800/Geraete/gs_200.pdf` last visited on 06.05.2026

[3] `https://www.3bscientific.com/product-manual/UE1070530_de.pdf` last visited on 14.06.2026

5 Appendix

5.1 Appendix A: Sound velocity in polyacryl

l [mm]	f [MHz]	Δt [μs]	v_{poly} [km/s]
41.30 ± 0.10	1	(32.0 ± 1.0)	(2.581 ± 0.081)
	2	(31.5 ± 1.0)	(2.622 ± 0.083)
	4	(30.0 ± 1.0)	(2.753 ± 0.092)
80.81 ± 0.10	1	(61.0 ± 1.0)	(2.650 ± 0.044)
	2	(60.5 ± 1.0)	(2.694 ± 0.045)
	4	(59.0 ± 1.0)	(2.739 ± 0.047)
120.76 ± 0.10	1	(92.0 ± 1.0)	(2.625 ± 0.029)
	2	(89.0 ± 1.0)	(2.714 ± 0.031)

Table 1: Data used for Experiment part 3.1, including lengths, frequencies, timedifferenze and corresponding sound velocity